常見兒童呼吸道疾病

兒童過敏氣喘風濕科
陳力振

Occurrence and relative importance of viruses in childhood respiratory tract infections

	Rhino-	RSV	PIV	Influenza-	Corona-	Adeno-	Entero-
No. of serotypes	>100	2	4	3	2	47	>70
Occurrence	All year	Epidemic	All year	Epidemic	All year	Sporadic	Sporadic
Common cold	++++	++	+	+	+++	+	++
Pharyngitis	+	+	++	++	++	++++	+++
Croup	+	++	++++	+++	+	++	+
Bronchiolitis		++++	++	+			+
Bronchitis	++++	++++	+++	++	++	+	+
Pneumonia	+++	++++	++	+	+	++	+

PIV : Parainfluenza viruses

Table 1. Viruses Detected in Nasopharyngeal Secretions from Hospitalized Children with Bronchiolitis.*

Virus	Type	Approximate Frequency %	Seasonality in North America
Respiratory syncytial virus	A and B	50–80	November through April
Human rhinovirus	Groups A, B, and C; >100 serotypes	5–25	Peak activity in spring and autumn
Parainfluenza virus	Type 3 most common, followed by types 1, 2, and 4	5–25	Type 3 is most prominent during spring, summer, and fall in odd-numbered years
Human metapneumovirus	Subgroups A and B	5–10	Late winter and early spring; season typically peaks 1–2 mo later than RSV peak
Coronavirus	OC43, 229E NL63, and HKU1	5–10	Winter and spring
Adenovirus	>50 serotypes	5–10	Year-round, although season for certain serotypes may be more restricted
Influenza virus	A and B	1–5	November through April
Enterovirus	Echovirus and coxsackievirus	1–5	Generally June through October

N Engl J Med 2016;374:62-72.

Airway Emergencies



Croup

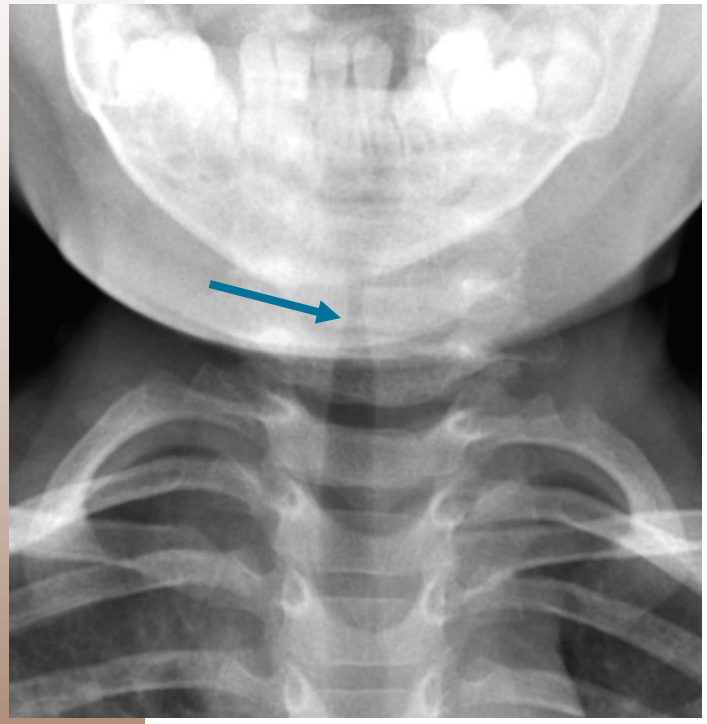


Epiglottitis



Foreign Body

Croup (Laryngotracheobronchitis)



Frontal radiograph shows straightening of the normally convex subglottic airway with narrowing "steeple sign"

- Clinical: Stridor, barking cough, fever
  - Peak incidence 6 mo to 3 yr
- Etiology: Viral
- Imaging
  - Frontal radiograph: loss of the normal shoulders at subglottic airway with narrowing
- Differentials
  - Normal epiglottis and prevertebral space helps in excluding epiglottitis and retropharyngeal abscess

Lab test & Imaging

Labs: Routine lab testing (*including* respiratory viral testing) **NOT recommended**

- Consider arterial blood gas if suspected impending respiratory failure

Imaging: Routine imaging CXR or lateral neck x ray **NOT recommended**

- Consider imaging if atypical presentation or suspected alternative diagnosis

Pharmacotherapy of Croup

Medication	Usual Dose	Dosing Range	Onset	Duration	Notes
Corticosteroids					
Dexamethasone	Preferred: 0.6 mg/kg (max 16 mg/dose) PO x1 Alternative: 0.6 mg/kg (max 16 mg) IM/IV x1	0.15-0.6 mg/kg (max 16 mg)	30 min	24-72 hours	▪ Repeat doses are NOT recommended
Budesonide	2 mg nebulized x1	0.5-4 mg (max 4 mg)	1-2 hours	-	▪ Consider as second-line to dexamethasone ▪ Repeat doses are NOT recommended ▪ Non-formulary at some institutions
Nebulized Epinephrine					
Racemic Epinephrine 2.25%	0.5 mL, diluted in 2 mL NS, nebulized q20 min PRN stridor <5 kg: consider 0.25 mL/dose	0.05-0.1 mL/kg (max 0.5 mL)	10-30 min	2-3 hours	▪ Usually use fixed dose ▪ Only for moderate severity and above ▪ Inpatient frequency: q2 hours PRN stridor
L-epinephrine 1:1000	0.5 mL/kg (max 5 mL/dose), diluted in 2 mL NS, nebulized q20 min PRN stridor	0.5 mL/kg (max 5 mL/kg)	10-30 min	2-3 hours	▪ Consider as alternative to racemic epinephrine if available in EMS vehicles ▪ Only for moderate severity and above ▪ Not available at many institutions

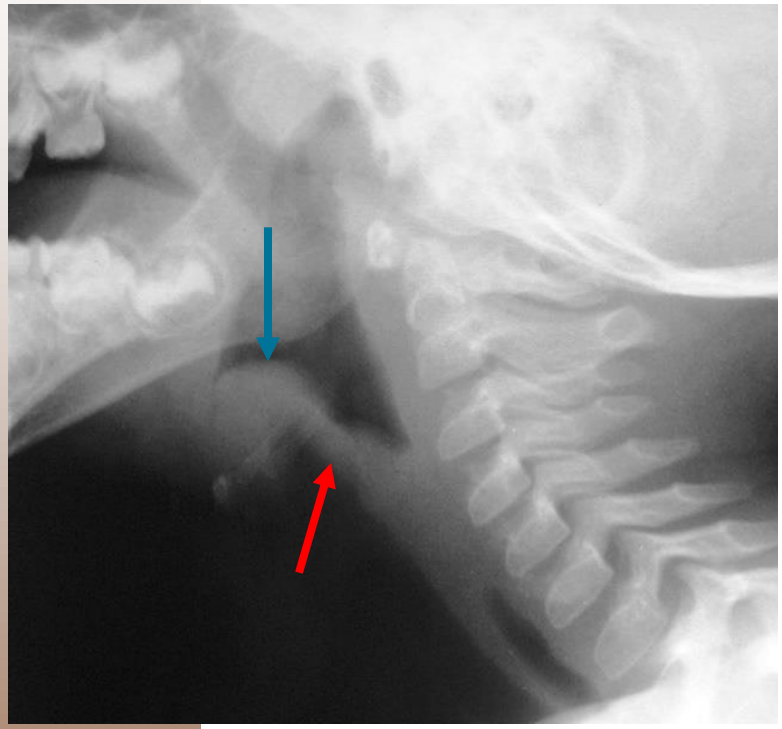
Admission Criteria

- Pediatric Ward
  - Persistent moderate symptoms after Dexamethasone & racemic epinephrine
  - Continued stridor at rest despite therapy
  - Inadequate hydration
  - Moderate work of breathing
  - Need for supplemental oxygen
  - Atypical presentation / concern for alternative diagnosis

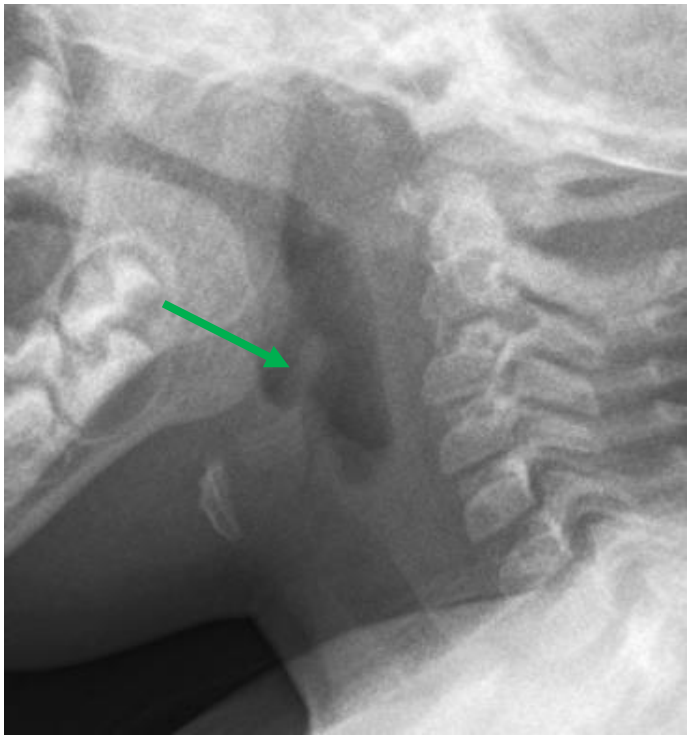
Admission Criteria

- PICU / Transfer to higher level of care
  - Persistent severe croup symptoms despite therapy
  - Escalating stridor at rest despite therapy
  - Need for intubation in ER
  - Impending respiratory failure
    - Bradypnea or poor respiratory effort
    - Severe work of breathing
    - Stridor may be present or decreased
    - Cyanosis / hypoxemia despite supplemental oxygen
    - Hypercarbia
    - Listlessness / decreased LOC

Epiglottitis



Lateral radiograph of the neck demonstrates thickening of the epiglottis (“thumb sign”) and aryepiglottic folds



Normal epiglottis for comparison

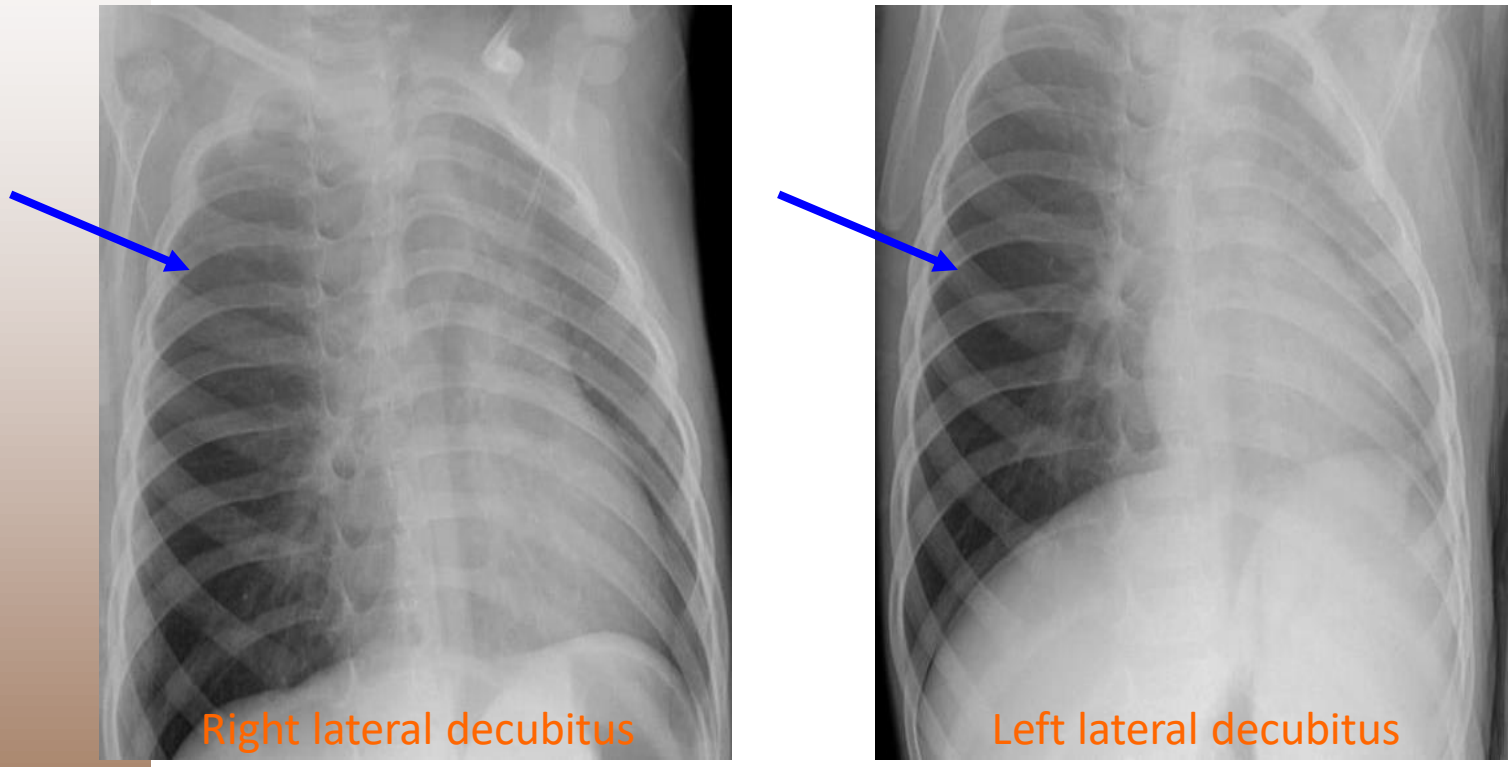
Epiglottitis

- Clinical: Fever, stridor, dysphagia, respiratory distress
- Etiology:
  - Most common pathogen is *H. influenzae* (decreasing incidence du to vaccination)
- Imaging
  - Lateral radiograph shows enlargement of the epiglottis and aryepiglottic folds
- Differential
  - Croup: diffuse narrowing of the subglottic airway
  - Retropharyngeal abscess: widening of the prevertebral soft tissue

Foreign body aspiration



Foreign Body Aspiration

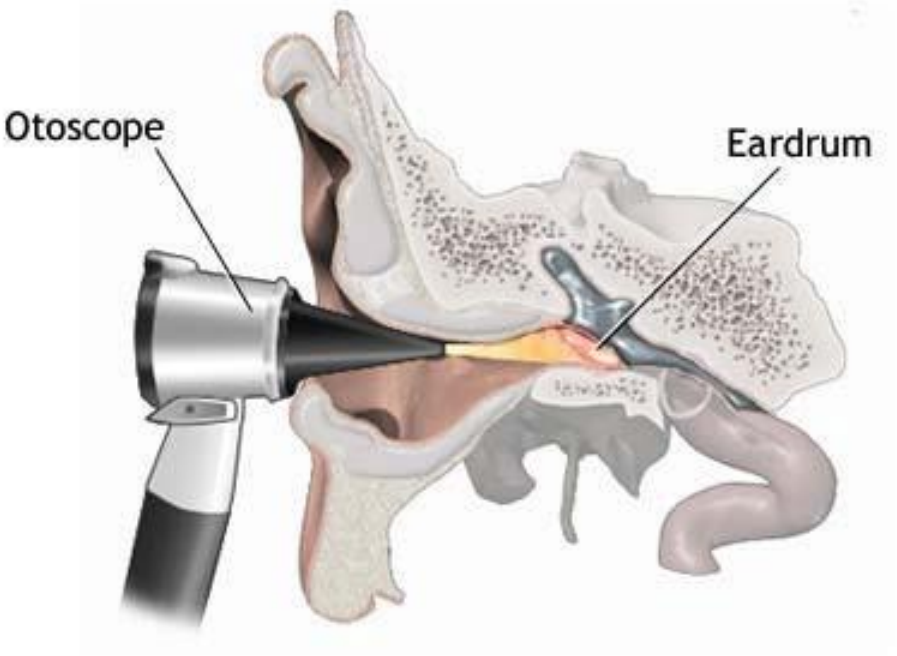


Lateral decubitus chest radiographs of a patient with confirmed right main stem bronchus foreign body demonstrate persistent asymmetric lucency of the right lung consistent with air trapping

Foreign body aspiration

- **Clinical:** Cough, respiratory distress
- **Etiology:**
  - Most common at ages 1-3 yr
  - Typically lodges in the right main stem bronchus
- **Imaging:**
  - Majority (80%) of aspirated FB radiolucent -> CXR may appear normal
  - Air trapping may be seen on expiratory or lateral decubitus views
- **Differentials:**
  - Asthma and viral respiratory tract infections typically result in symmetric lu volumes
  - Congenital lobar emphysema: confined to a single lobe

Acute Otitis Media





A.D.A.M.

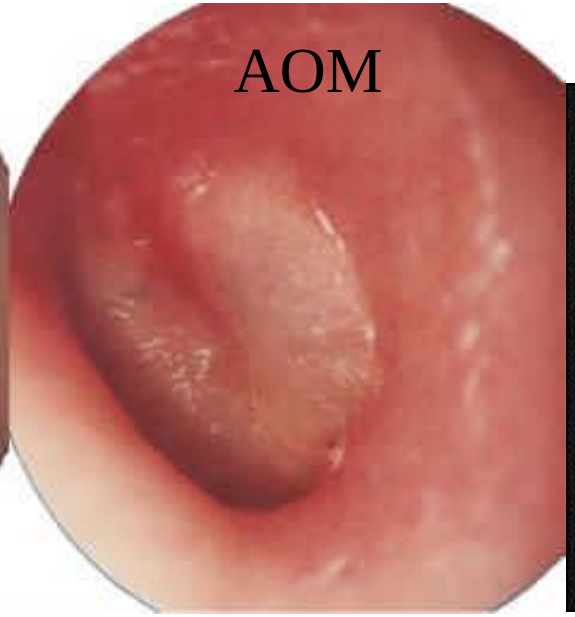


Hemorrhagic,translucent, mobile TM with URI

Normal



AOM



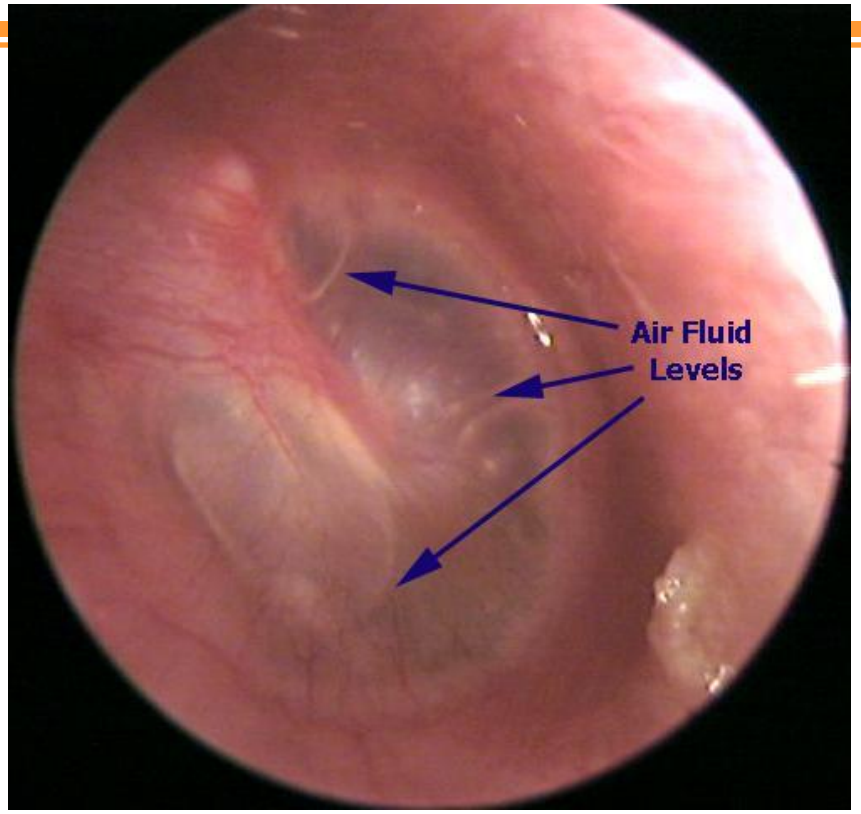


Position, mobility, color and degree of translucency

The redness of TM alone does not suggest the diagnosis of AOM

OME





serous fluid & a normal light

Definition of AOM

- **Diagnosis of AOM**
  - **1) a history of acute onset of signs & symptoms**
  - **2) the presence of MEE, and**
  - **3) signs or symptoms of middle-ear inflammation**

1. **MEE that is indicated by any of the following:**
   a. **Bulging of the tympanic membrane**
   b. **Limited or absent mobility of the tympanic membrane**
   c. **Air-fluid level behind TM**
   d. **Otorrhea**
2. **Signs or symptoms of middle-ear inflammation as indicated by either**
   a. **erythema or**
   b. **otalgia (interference normal activity/sleep)**

Pediatrics 2004;113;1451-65

Acute Otitis Media

- Acute otitis media (AOM) is a common condition for which antibiotics are frequently prescribed
  - In the UK 30% of children under 3 years of age visit their general practitioner with AOM each year, and 97% receive antimicrobial treatment. 10% of children have had an episode of AOM by the age of 3 months
- Diagnosing AOM based on signs and symptoms is not clear-cut but ear pain, a cloudy, bulging or red tympanic membrane may help with diagnosis
- The most common bacterial cause of otitis media are: *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*.
- **60% of children will improve within 24 hours and 80% resolve in about 3 days without antibiotic treatment**
- Serious complications are rare in otherwise healthy children but include hearing loss, mastoiditis, meningitis, and recurrent attacks

Clinical Evidence. Neill O, et al. Search date Jan 2006

AAP Guideline for Rx of AOM

Age of child	If diagnosis is certain	If diagnosis is uncertain
< 6 m	Antibiotic	Antibiotic
6 m- 2 yrs	Antibiotic	Antibiotic if severe, observe if nonsevere illness
≥ 2 yrs	Antibiotic if severe, observe if non severe	Observe

Treatment of AOM

- **Analgesia.** Paracetamol (NNT = 6) and ibuprofen (NNT=5) are effective in relieving ear ache but not other symptoms of AOM (appearance of the tympanic membrane; rectal temperature; and parental assessment of appetite, sleep, and playing activity)

- **Antibiotics**.
  - 7 patients in 100 will gain extra pain relief at 2–7 days with antibiotics.
  - There is evidence that younger patients (under 2 years) may benefit more. In the same patient group, 6 extra children will experience harms such as diarrhoea, vomiting or rash
  - It is not possible to identify which patients will gain extra pain relief and which will suffer increased harms from antibiotic treatment

- Antibiotics had no effect on pain outcomes at 24 hours nor the rate of subsequent recurrence of AOM or abnormal tympanometry at 1 month or at 3 months

Clinical Evidence. Neill O, et al. Search date Jan 2006

10 Warning Signs of Primary Immunodeficiency

Primary Immunodeficiency (PI) causes children and adults to have infections that come back frequently or are unusually hard to cure. 1:500 persons are affected by one of the known Primary Immunodeficiencies. If you or someone you know is affected by two or more of the following Warning Signs, speak to a physician about the possible presence of an underlying Primary Immunodeficiency.

1. 
Four or more new ear infections within one year.

2. 

Two or more serious sinus infections within one year.

3. 

Two or more months on antibiotics with little effect.

4. 
Two or more pneumonias within one year.

5. 

Failure of an infant to gain weight or grow normally.

6. 
Recurrent, deep skin or organ abscesses.

7. 
Persistent thrush in mouth or fungal infection on skin.

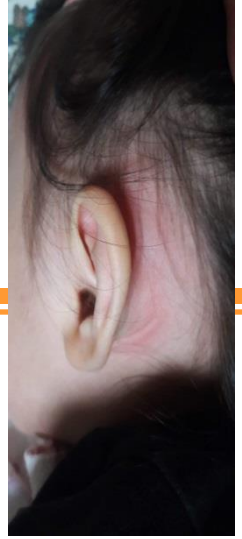
8. 
Need for intravenous antibiotics to clear infections.

9. 

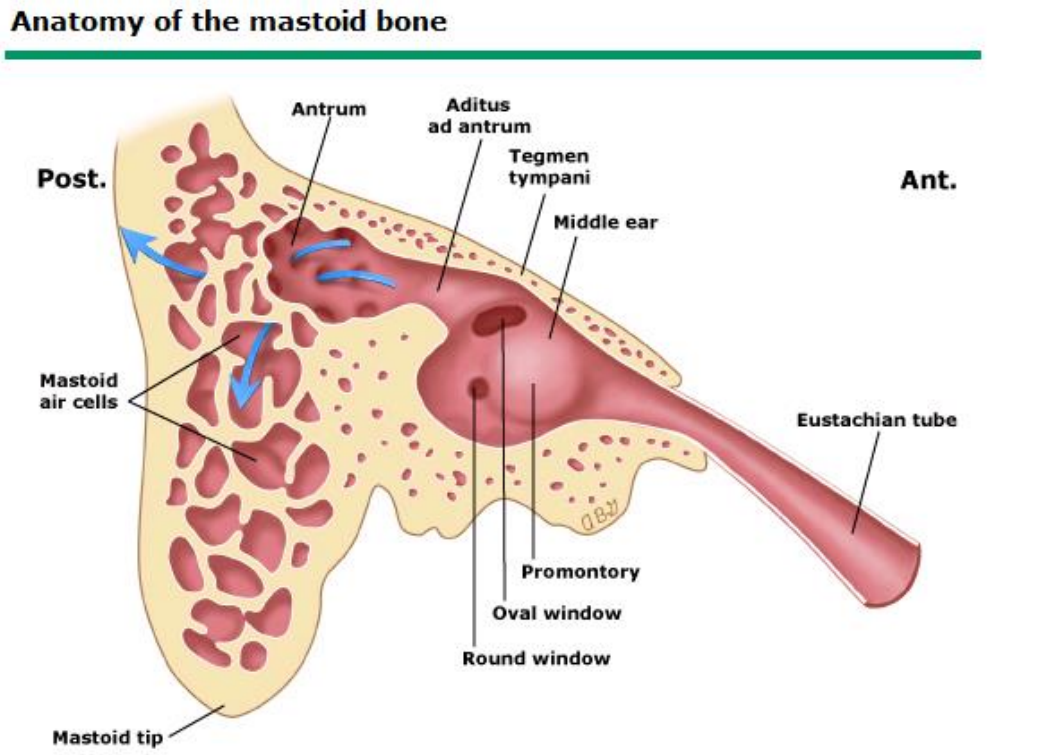
Two or more deep-seated infections including septicemia.

Mastoiditis

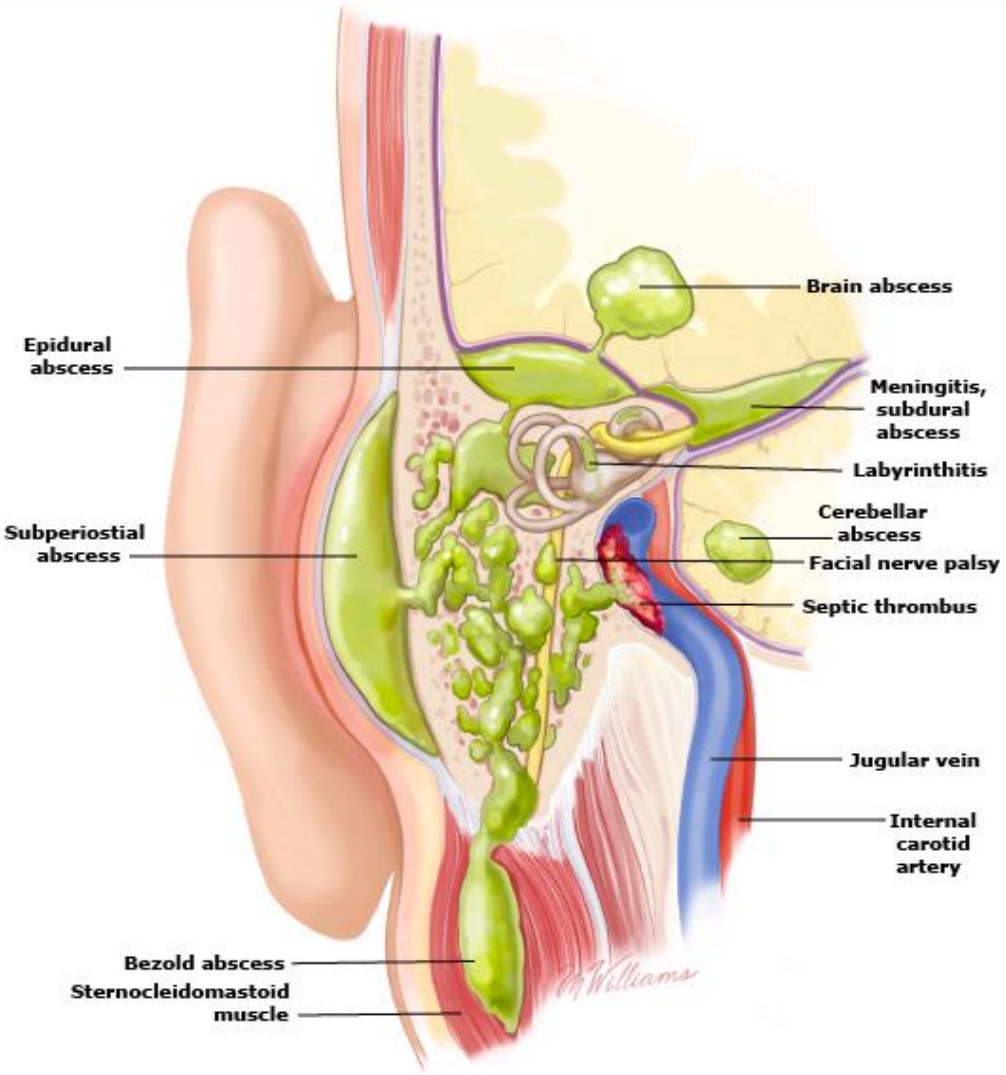
乳突炎







Complications of acute mastoiditis



Spread of infection from the mastoid to adjacent areas may lead to intratemporal and intracranial complications as depicted above.

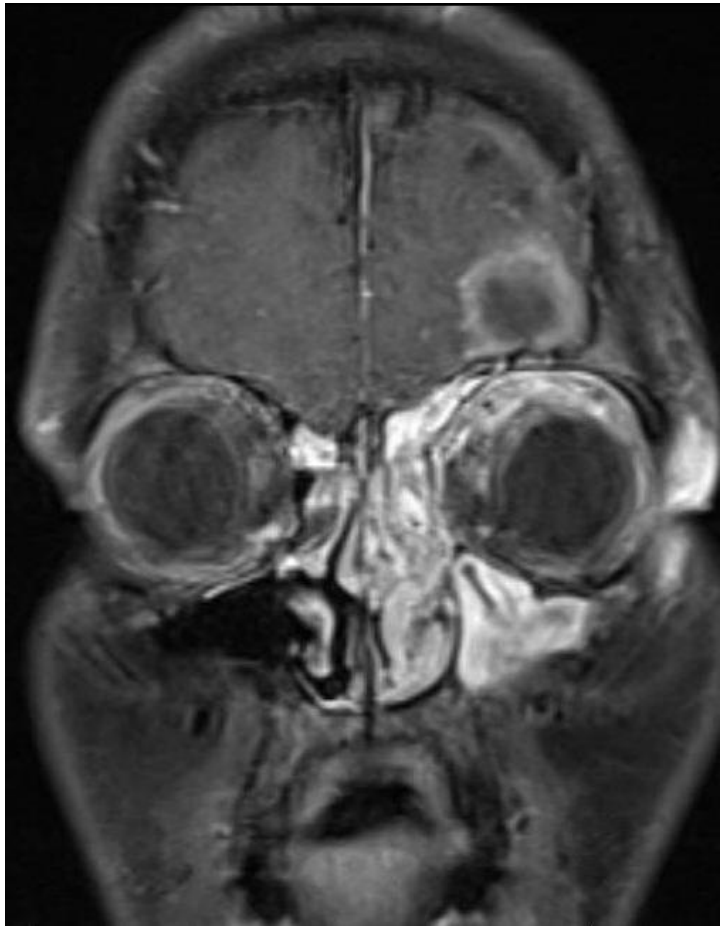
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Sinusitis

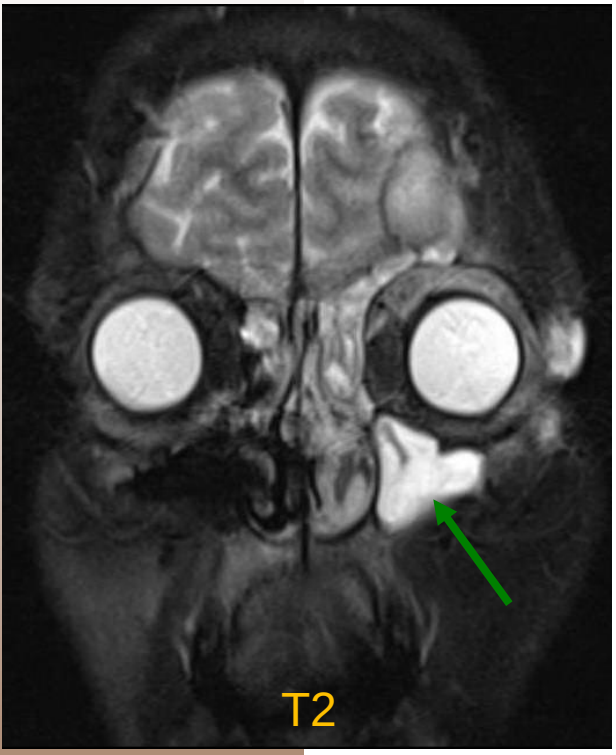
Home

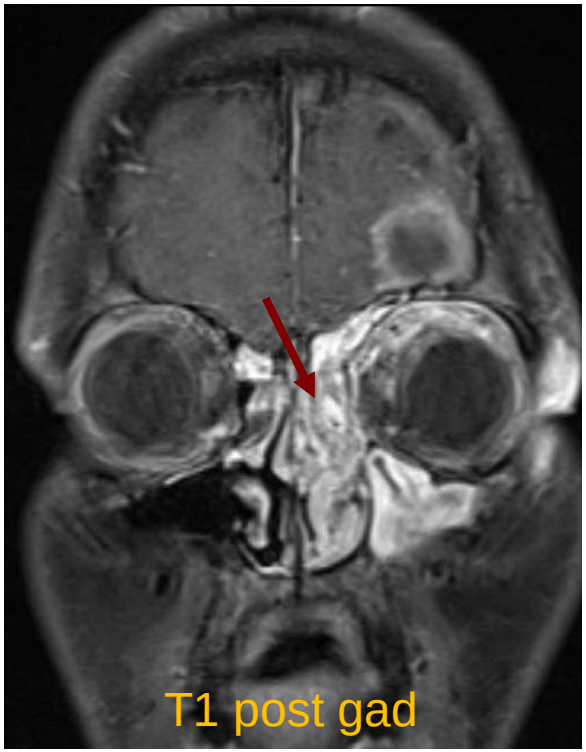
- Clinical: URI symptoms, nasal discharge, fever
- Complications: Orbital cellulitis, osteomyelitis, intracranial abscess,
- Imaging: Mucosal thickening/enhancement, sinus opacification, air-fluid levels
  - Look for intracranial and intraorbital extension for early complication detection

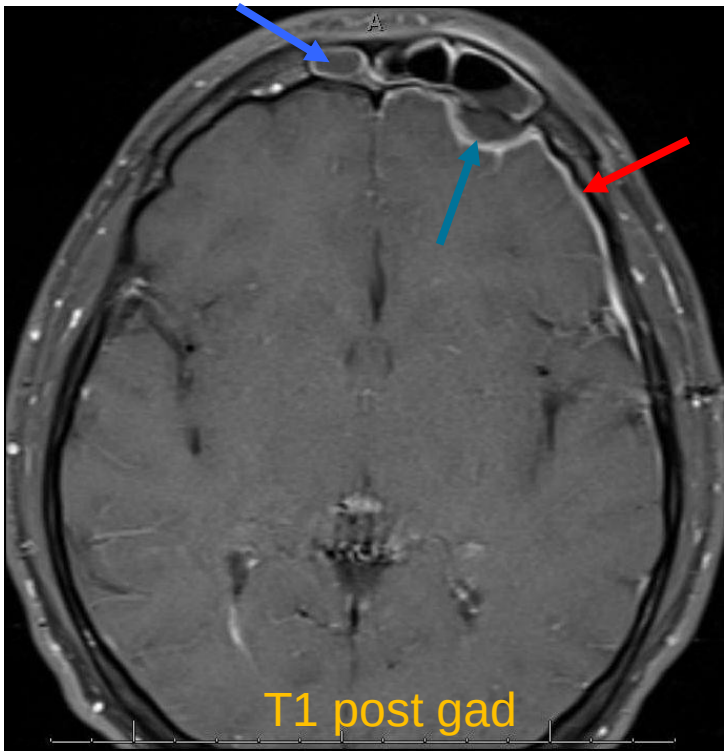


Home

Sinusitis with Epidural Abscess







MR images show fluid signal intensity and enhancement in the left maxillary, ethmoid, and frontal sinuses with underlying pachymeningeal enhancement and rim-enhancing epidural abscess along the anterior frontal convexity

Acute Bronchiolitis

Indications for Hospitalization in Bronchiolitis Caused by RSV

- Age < 3 months
- Gestational age at birth of less than 34 weeks
- Cardiopulmonary disease or immuodeficiency
- RR > 70/min
- Lethargic appearance
- Wheezing and respiratory distress associated with SaO2 < 92 percent on room air
- Hypercarbia
- Atelectasis or consolidation on chest radiography

Table 2. American Academy of Pediatrics Guidance for Diagnosis and Management of Bronchiolitis.*

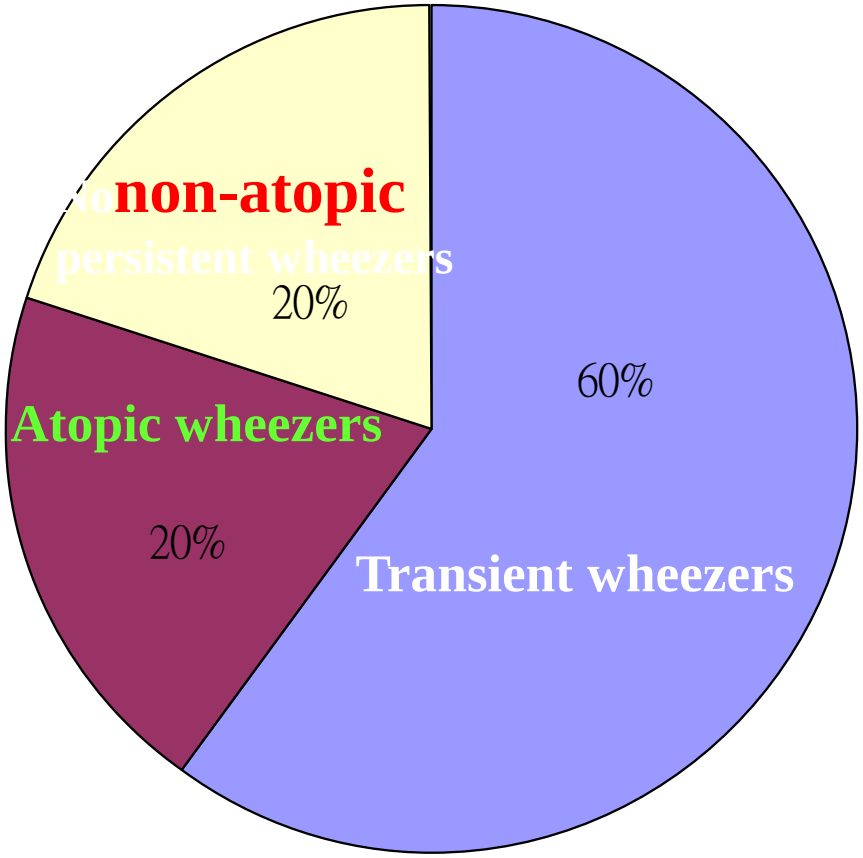
Intervention	Recommendation	Comment
Diagnostic Test		
Chest radiography	Not recommended for routine use	Poor correlation with severity of disease or risk of progression; studies show increase in inappropriate use of antimicrobial therapy owing to similar radiographic appearance of atelectasis and infiltrate
Testing for viral cause	Not recommended for routine use	May influence isolation of symptomatic patients, but infection-control procedures are similar for most respiratory viruses
Treatment		
Bronchodilator therapy	Not recommended	Randomized trials have not shown a consistent beneficial effect on disease resolution, need for hospitalization, or length of stay
Epinephrine	Not recommended	Large, multicenter, randomized trials have not shown improvement in outcome among outpatients with bronchiolitis or hospitalized children
Glucocorticoid therapy	Not recommended	Large, multicenter, randomized trials provide clear evidence of lack of benefit
Nebulized hypertonic saline	May be considered	Nebulized 3% saline may improve symptoms of mild-to-moderate bronchiolitis if length of stay is >3 days (most hospitalizations are <72 hr)
Supplemental oxygen	Routine use not recommended if oxyhemoglobin saturation is >90% in the absence of acidosis	Transient episodes of hypoxemia are not associated with complications; such episodes occur commonly in healthy children
Pulse oximetry	Not recommended for patients who do not require supplemental oxygen or if oxygen saturation is >90%	Oxygen saturation is a poor predictor of respiratory distress; routine use correlates with prolonged stays in the emergency department and hospital
Chest physiotherapy	Not recommended	Deep suctioning is associated with a prolonged hospital stay; removal of obstructive secretions by suctioning the nasopharynx may provide temporary relief
Antimicrobial therapy	Not recommended for routine use	Risk of serious bacterial infection is low; routine screening is not warranted, especially among infants 30 to 90 days of age
Nutrition and hydration	Hospitalization for observation of hydration and nutritional status may be needed for infants with respiratory distress	Intravenous or nasogastric hydration may be used

N Engl J Med 2016;374:62-72

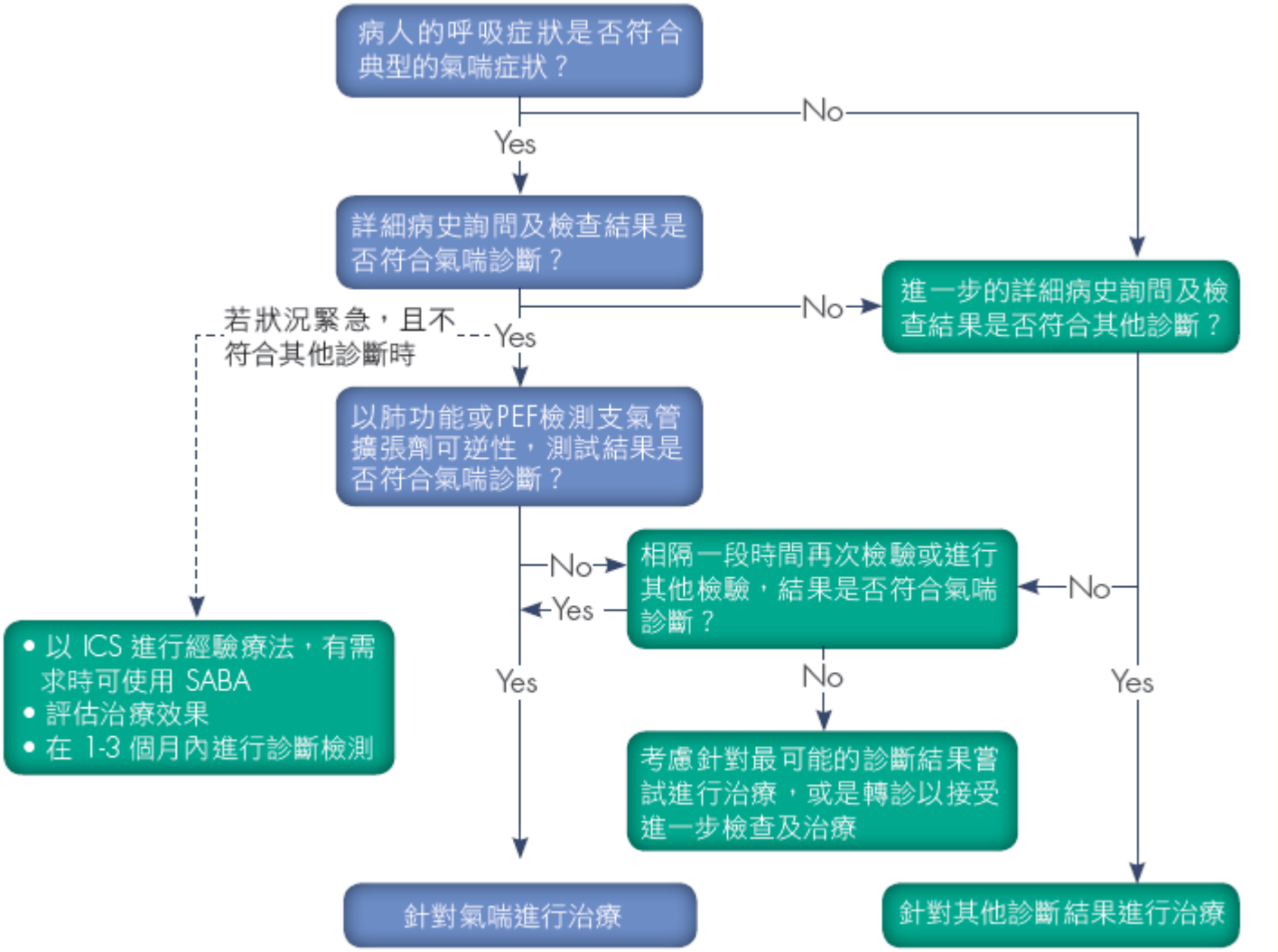
Pathophysiology of wheezing

- Bronchospasm
- Mucosal edema
- Mucus plugging
- Airway geometry and mechanics
- ↓ elastic recoil

Wheezing phenotypes in infants

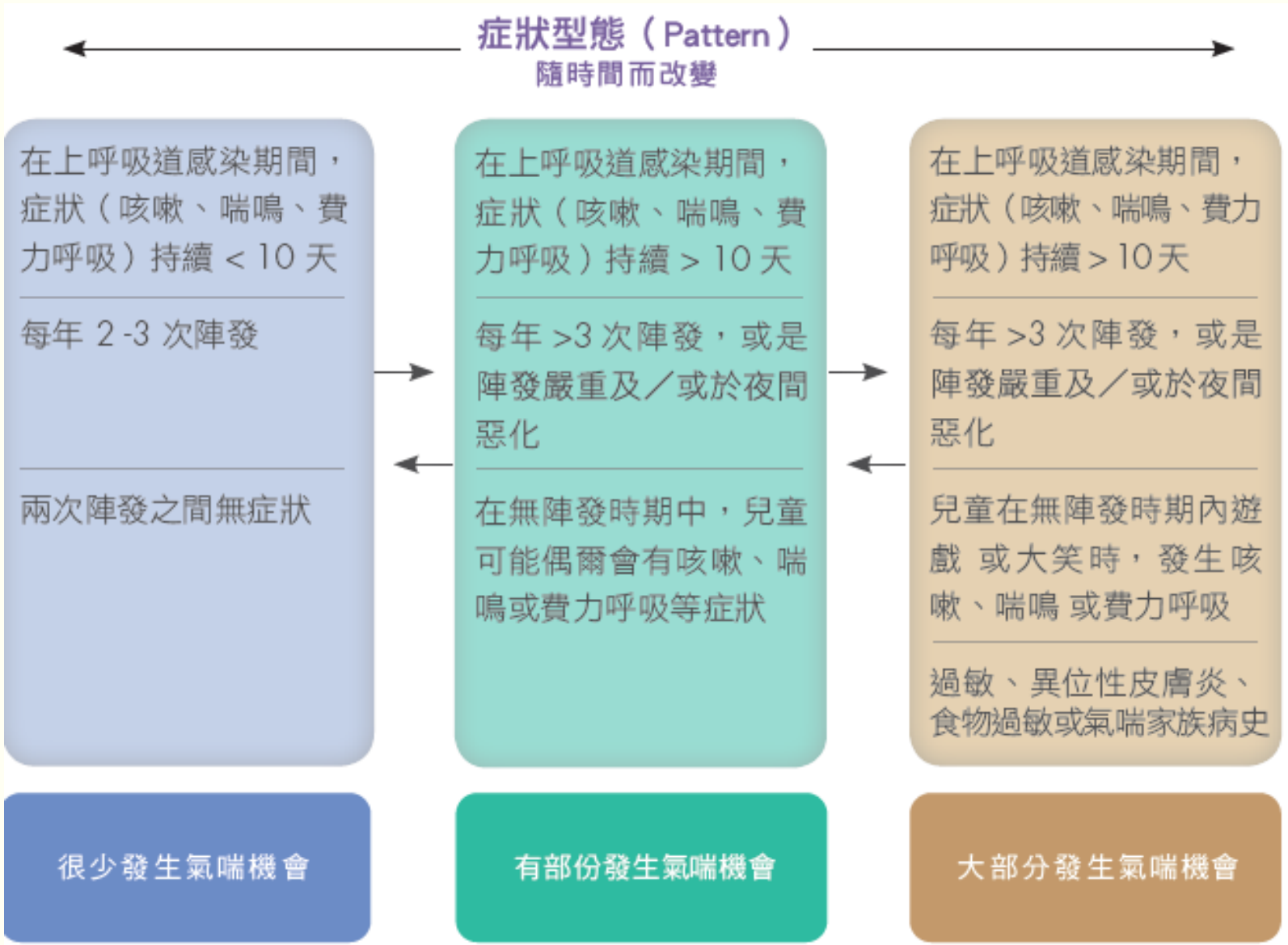


氣喘臨床診斷流程圖



ICS：吸入性類固醇。PEF：尖峰呼氣流速。SABA：短效乙二型支氣管擴張劑

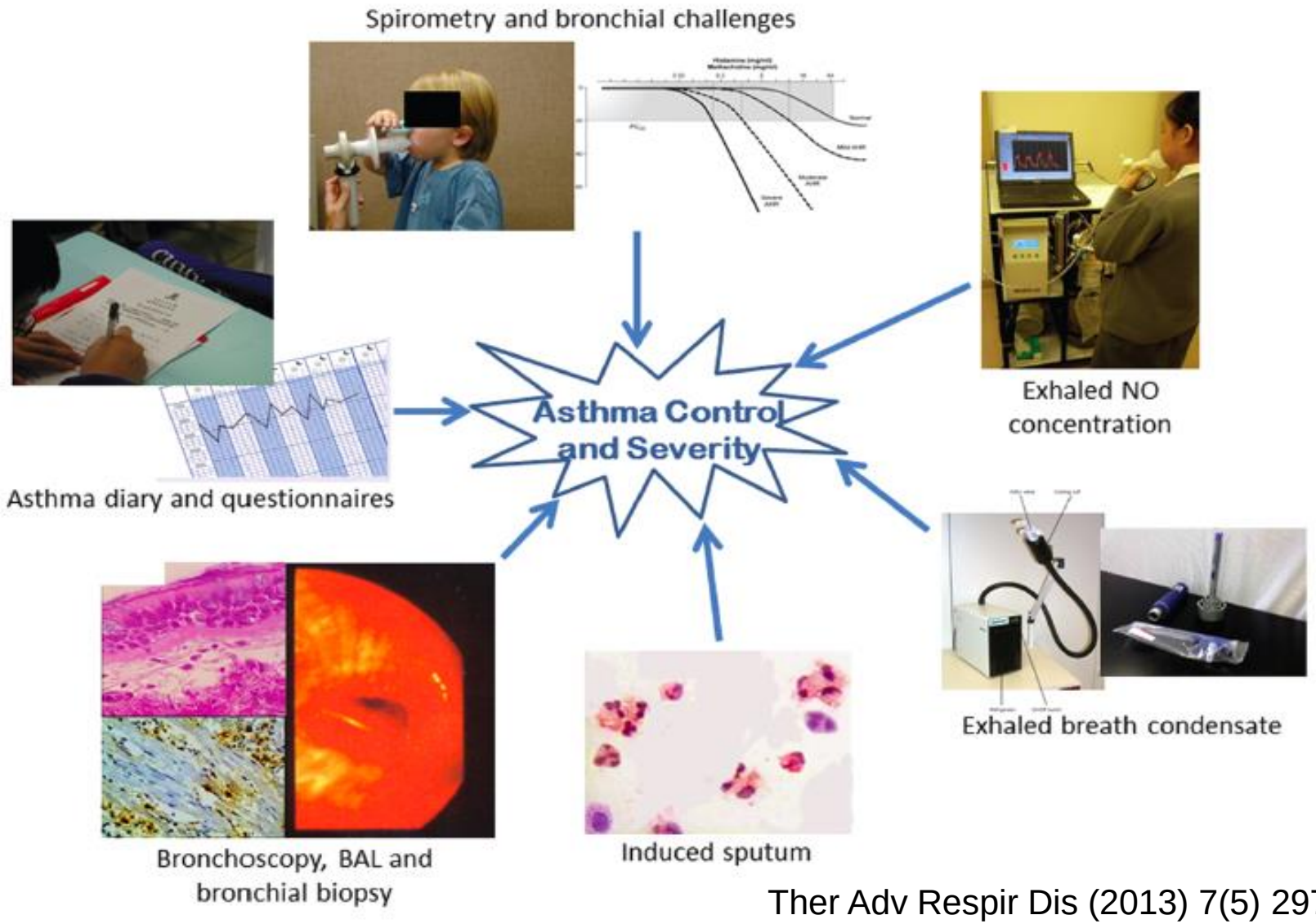
5歲以下喘鳴兒童的氣喘診斷可能性



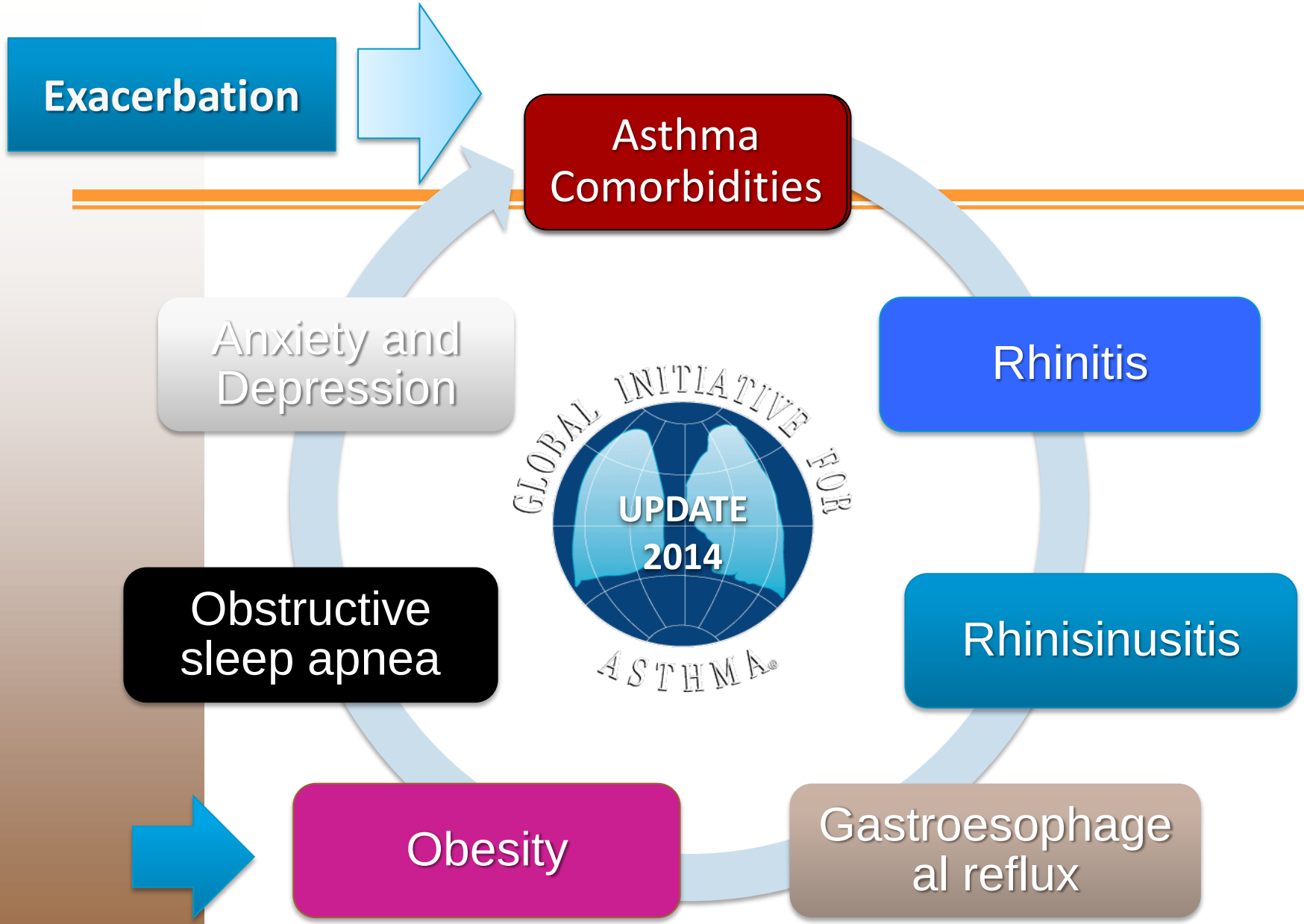
5歲以下兒童中氣喘常見的鑑別診斷

病症	典型特徵
反覆病毒性呼吸道感染	主要為咳嗽、流鼻水、鼻塞，持續 10 天。在兩次感染之間無症狀。
胃食道逆流	進食時咳嗽、反覆性胸腔感染、容易嘔吐（尤其是大量進食時）、對氣喘藥物反應不佳。
吸入異物	在進食或遊戲期間突然發生嚴重的咳嗽或喘鳴、反覆性胸腔感染和咳嗽、局部肺部徵象。
氣管軟骨軟化	哭泣、進食或上呼吸道感染期間的呼吸存在雜音、咳嗽聲尖銳、吸氣或呼氣時肋間凹陷。症狀通常在出生時即出現，對於氣喘藥物反應不良。
結核	持續性伴隨痰音的咳嗽、對於一般抗生素無反應的發燒、淋巴結腫大、對支氣管擴張劑或ICS反應不良、曾接觸肺結核病人。
先天性心臟病	心雜音、進食時發紺、生長停滯、心跳過快、呼吸過快或肝腫大、對氣喘藥物反應不良。
纖維囊腫(cystic fibrosis)	出生時就後不久即開始咳嗽、反覆性胸腔感染、生長停滯（吸收不良），且糞便結構鬆散、油膩、體積龐大。
原發性絨毛運動障礙 (primary ciliary dyskinesia)	咳嗽和反覆性輕微胸腔感染、慢性耳部感染和鼻分泌物帶膿、對氣喘藥物反應不良。大約 50% 患有此病症的兒童為器官轉位者 (situs inversus)。
血管環	呼吸通常持續帶有雜音、對氣喘藥物反應不良。
肺支氣管發育不全 (bronchopulmonary dysplasia)	嬰兒早產、出生體重極低、需要長時間使用呼吸機換氣或補充氧氣、出生後即開始有呼吸困難。
免疫不全	反覆發燒和感染（包括非呼吸道感染）、生長停滯。
細菌性支氣管炎	持續性濕咳，對氣喘藥物的反應不良

Asthma Status Assessment

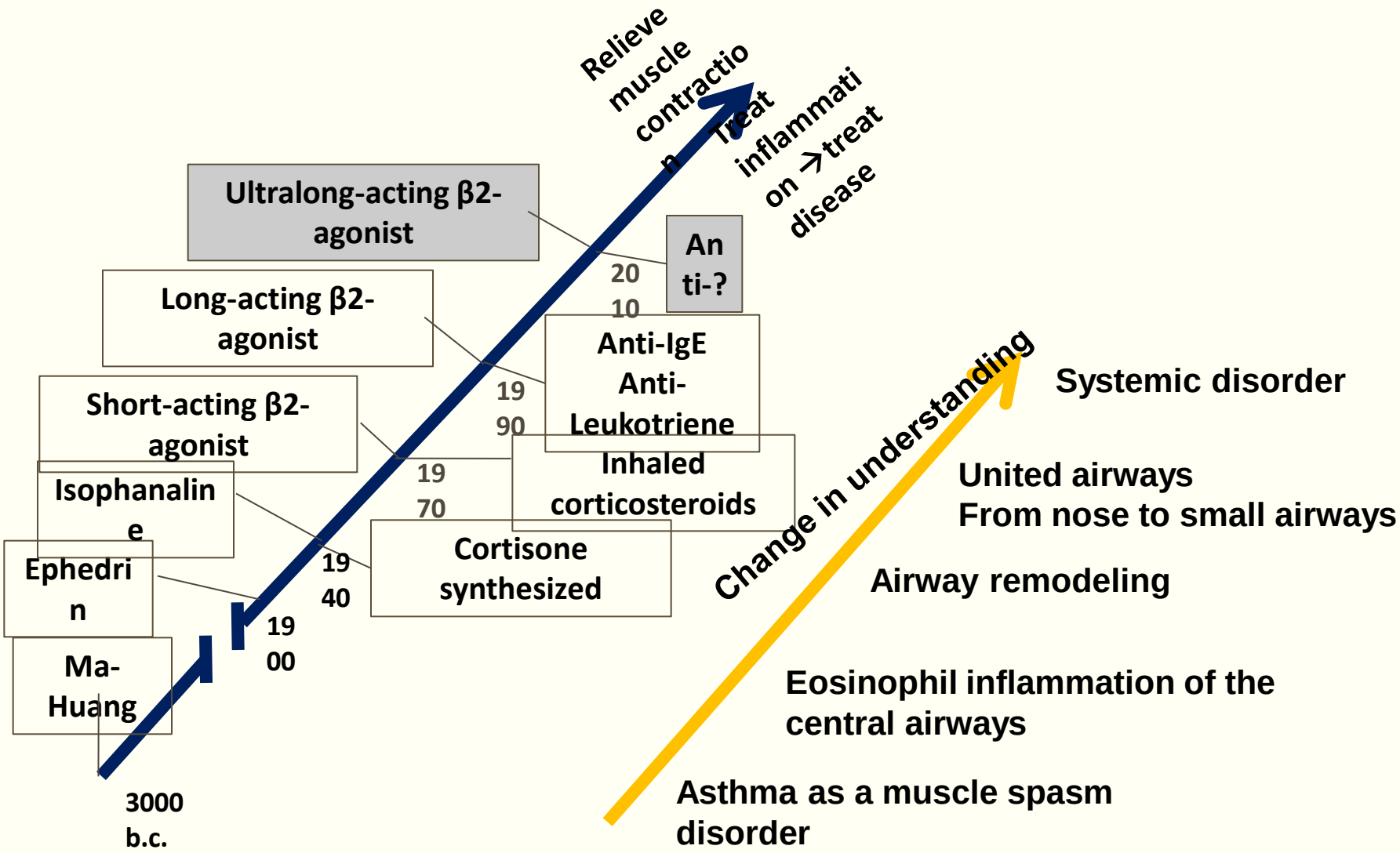


Ther Adv Respir Dis (2013) 7(5) 297–308



GINA 2014 – May contribute to symptoms, poor quality of life and poor asthma control

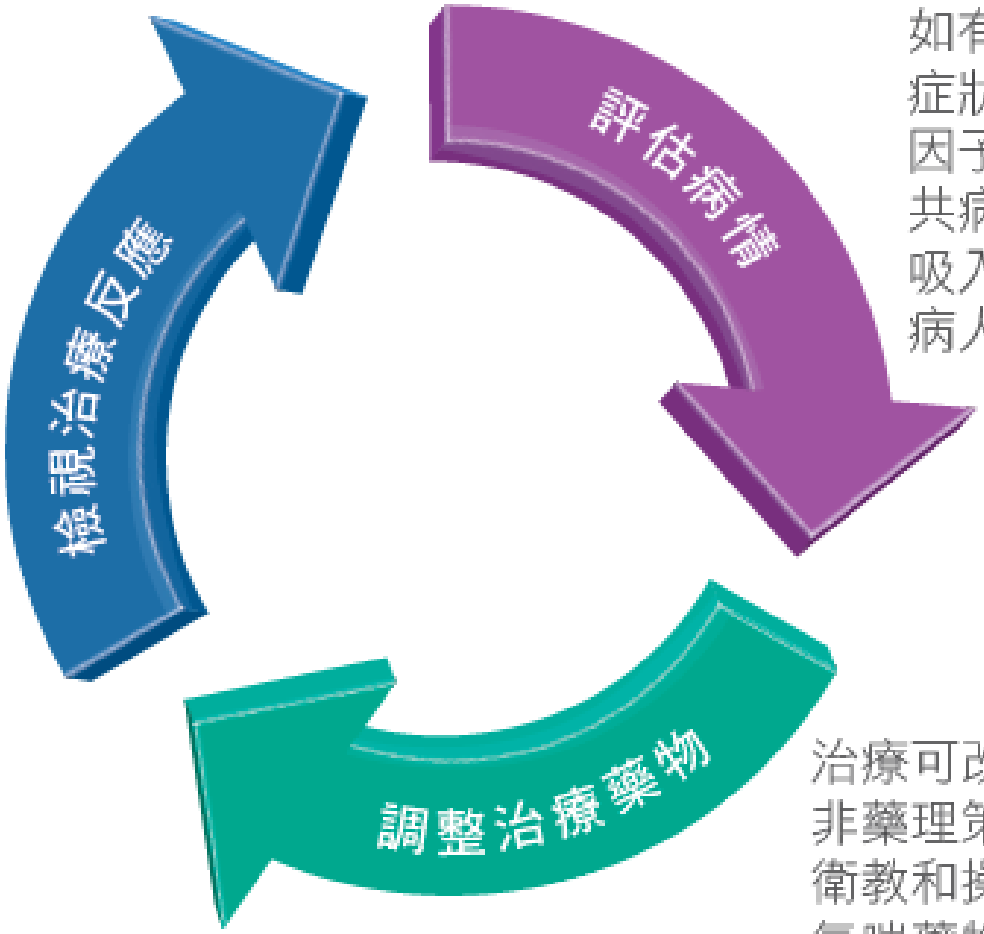
Evolution of asthma treatment



氣喘之藥物治療

- **治本**：抗發炎藥物(anti-inflammatory agents)─
  - 類固醇(glucocorticosteroid)
    - 吸入型、口服型、或注射型
  - 肥大細胞安定劑(mast cell stabilizer)─
    - 色甘酸(cromolyn)、奈多羅米(nedocromil)
  - 白三烯拮抗劑(leukotriene antagonists)─
    - Montelukast, zafirlukast
  - 抗免疫球蛋白 E (anti-Ig E therapy)─**Omalizumab**
- **治標**：支氣管擴張劑(bronchodilators)─
  - 乙二型交感神經興奮劑(β_2 agonists)─
    - 短效(SABA)與長效(LABA)
  - 抗膽鹼藥物(anticholinergics)─
    - 短效(SAMA)與長效(LAMA)
  - 茶鹼(xanthines)
  - 其他 : MgSO4,氦氧混合物(Heliox)

以氣喘控制為導向之氣喘治療的三大循環要點

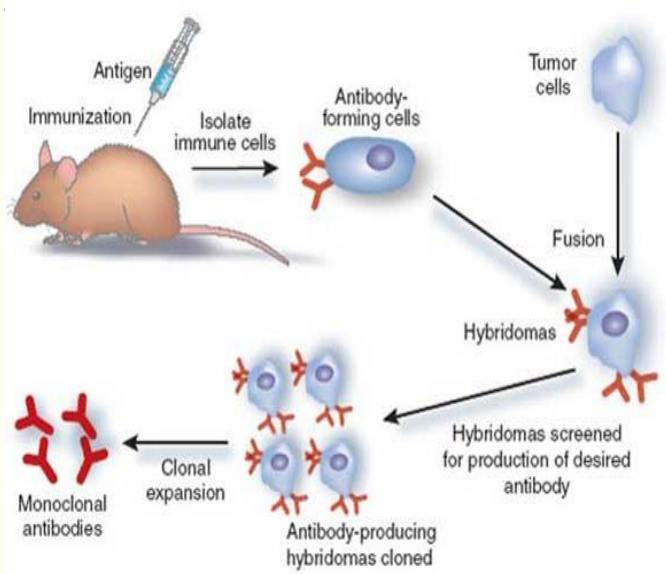


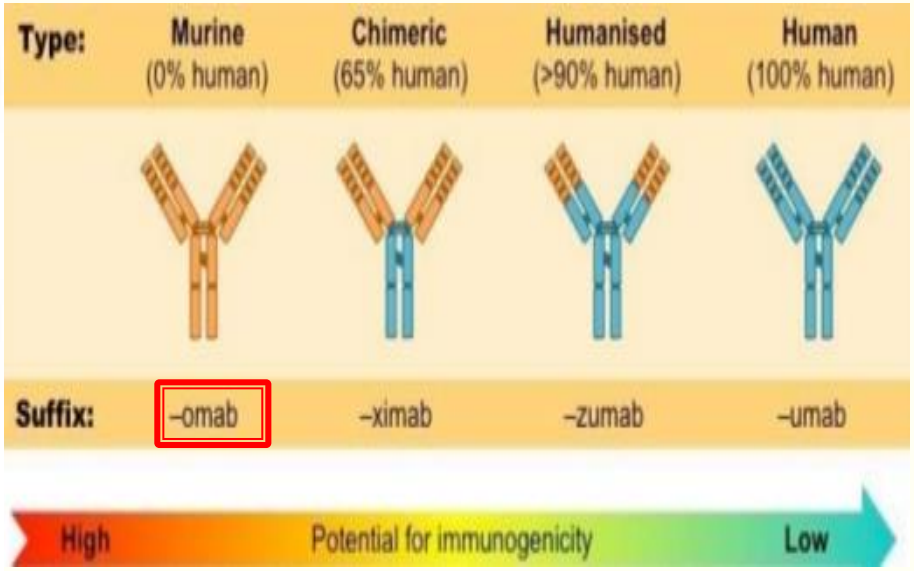
如有需要先確認診斷
症狀控制和可改善的危險因子(包含肺功能)
共病
吸入器的使用技巧和遵囑性
病人(和家屬)的治療目標

治療可改善的危險因子和共病症
非藥理策略
衛教和操作技巧訓練
氣喘藥物

症狀
惡化
副作用
肺功能
病人(和家屬)的滿意度

Monoclonal Antibodies





Barone's Guide to Monoclonal Antibody Nomenclature

How to name a monoclonal antibody:

MABs are named by combining a **Prefix** + **Target** + **Source** + **Suffix** (usually "mab")

Prefix	+	Antibody Target		+	Antibody Source		+	Suffix
		• Bone	-o(s)-		• Human	-u-		-mab
		• Cardiovascular	-c(i)-		• Humanized	-zu-		
		• Immune-modulating	-l(i)-		• Chimeric	-xi-		
		• Interleukin	-k(i)-		• Rat	-a-		
		• Tumor	-t(u)-					
		• Virus	-v(i)-					

Ex: Oma-li-zumab

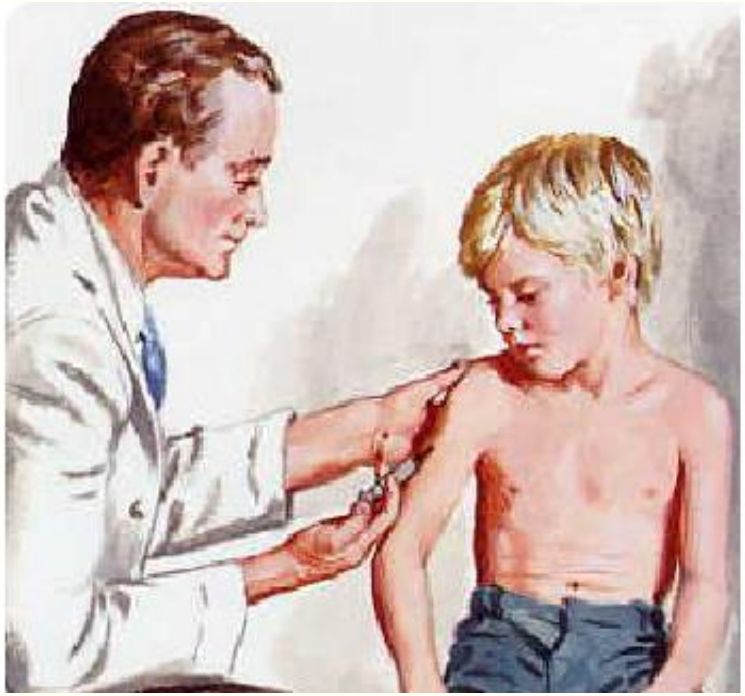
Biologic therapies for severe asthma

Biologic therapy	Target	Approval date/status
Omalizumab	IgE	2003
Mepolizumab	IL-5	2015
Reslizumab	IL-5	2016
Benralizumab	IL-5Rα	2017
Dupilumab	IL-4Rα (IL-4/IL-13)	2018
Tezepelumab	TSLP	Phase 2B

TSLP, thymic stromal lymphopoietin.

Curr Opin Allergy Clin Immunol 2018,

減敏治療給予路徑

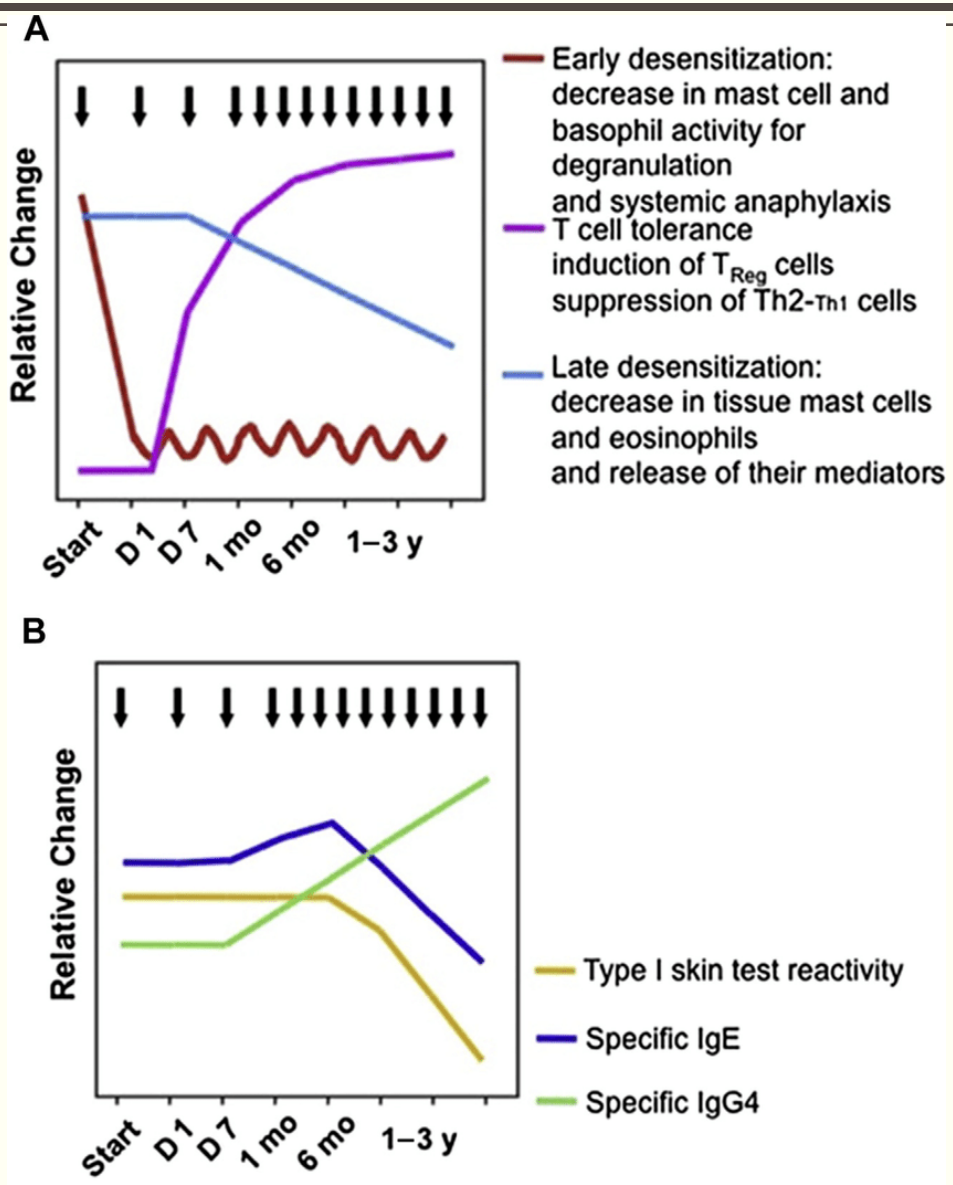


SCIT (Subcutaneous immunotherapy)
皮下注射



SLIT(Sublingular immunotherapy)
舌下給藥

Immunologic changes during allergen immunotherapy



Evaluate the efficacy of anti-asthma medication in asthma patients

- Lung function, FEV_1, PEF
- Acute Exacerbation
- Quality of life
- Exercise capability
- Airway responsiveness
- eNO reflects acute but not remodeling